Elementary Analysis

Chapter 2, Section 13

Some Topological Concepts in Metric Spaces
Exercises 13.1–13.15, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

13 Some topological concepts in metric spaces **3**

13.1 Exercise 13.1: Two complete metrics on finite-dimensional space 3
13.2 Exercise 13.2: Euclidean distance and convergence of coordinates 4
13.3 Exercise 13.3: The supremum metric on bounded sequences 5
13.4 Exercise 13.4: Unions and finite intersections of open sets 6
13.5 Exercise 13.5: De Morgan's law and intersections of closed sets 6
13.6 Exercise 13.6: Four equivalent descriptions involving closure 7
13.7 Exercise 13.7: Decomposing an open real set into disjoint intervals 8
13.8 Exercise 13.8: Intervals, balls, half-spaces, and examples that are neither 9
13.9 Exercise 13.9: Three closures in the real line 10
13.10 Exercise 13.10: Three sets with no interior points 11
13.11 Exercise 13.11: Compactness and subsequences in Euclidean space 12
13.12 Exercise 13.12: Closed subsets and finite unions of compact sets 13
13.13 Exercise 13.13: A compact real set contains its extreme bounds 13

13.14 Exercise 13.14: The diameter of a compact set is attained 14
13.15 Exercise 13.15: A closed bounded set that is not compact 15

13 Some topological concepts in metric spaces

Throughout this section, k is a positive integer. In $\mathbb{R}^k$, d without a subscript denotes Euclidean distance. In an arbitrary metric space (S, d), write

$$B(x, r) = \{y \in S : d(y, x) < r\} \qquad (r > 0).$$

The ambient space matters: complements, interiors, and closures are taken in the space specified in the exercise. We write $\overline{E}$ for the book's closure E^-, and E° for its interior.

We will use the following elementary ball fact. If $y \in B(x, r)$, then $\rho = r - d(y, x) > 0$. For $z \in B(y, \rho)$, the triangle inequality gives

$$d(z, x) \le d(z, y) + d(y, x) < \rho + d(y, x) = r.$$

Thus $B(y, \rho) \subseteq B(x, r)$, which proves that every open ball in every metric space is open. No assertion about the boundary of a ball in a general metric space is being assumed.

13.1 Exercise 13.1: Two complete metrics on finite-dimensional space

What we must show. For $x, y \in \mathbb{R}^k$, verify that

$$d_1(x, y) = \max_{1 \le j \le k} |x_j - y_j|, \qquad d_2(x, y) = \sum_{j=1}^{k} |x_j - y_j|$$

are metrics, and prove that each metric is complete.

Plan. Check the three metric axioms directly. For completeness, first obtain a limit of each coordinate, then combine the finitely many coordinate estimates using one cutoff.

(a), first axiom: finite nonnegative distances, with the correct zeros.

Each $|x_j - y_j|$ is a finite nonnegative real number. A finite maximum and a finite sum of these quantities are therefore finite and nonnegative. If $x = y$, every coordinate difference is zero, and both distances are zero. Conversely, if the maximum is zero, then every nonnegative coordinate difference is at most zero and so is zero. If the sum is zero, every nonnegative summand must also be zero. In either case $x_j = y_j$ for all j, hence $x = y$. Equivalently, if $x \neq y$, at least one difference is positive, and both distances are positive.

(a), second and third axioms: symmetry and triangle inequalities.

The identity $|x_j - y_j| = |y_j - x_j|$ gives symmetry for both formulas. For any $x, y, z \in \mathbb{R}^k$, the real triangle inequality gives

$$|x_j - z_j| \le |x_j - y_j| + |y_j - z_j|.$$

For d_1, the right side is at most $d_1(x, y) + d_1(y, z)$ for every j. Taking the maximum of the left sides proves $d_1(x, z) \le d_1(x, y) + d_1(y, z)$. For d_2, summing the coordinate inequalities gives

$$d_2(x, z) \le \sum_{j=1}^{k} |x_j - y_j| + \sum_{j=1}^{k} |y_j - z_j| = d_2(x, y) + d_2(y, z).$$

This checks all metric axioms.

(b), coordinate limits: start with a Cauchy sequence in either metric.

Let $(x^{(n)})$ be Cauchy for d_i, where $i = 1$ or $i = 2$. For every coordinate j,

$$\left|x_j^{(m)} - x_j^{(n)}\right| \leq d_i(x^{(m)}, x^{(n)}).$$

For d_1, this is the defining maximum bound; for d_2, it is the bound of one nonnegative summand by the whole sum. Given $\varepsilon > 0$, the Cauchy cutoff for d_i therefore also makes every coordinate difference smaller than ε. Each coordinate sequence is Cauchy in $\mathbb{R}$. Completeness of $\mathbb{R}$ gives a real limit x_j for each j. Put $x = (x_1, \ldots, x_k) \in \mathbb{R}^k$.

(b), convergence in the maximum metric.

Fix $\varepsilon > 0$. For each j, choose N_j so that $n > N_j$ implies $\left|x_j^{(n)} - x_j\right| < \varepsilon$. Take $N = \max\{N_1, \ldots, N_k\}$. If $n > N$, every one of the finitely many differences is strictly less than ε. Their maximum is therefore strictly less than ε: $d_1(x^{(n)}, x) < \varepsilon$.

(b), convergence in the sum metric.

For d_2, instead choose N_j to give coordinate errors smaller than ε/k, which is positive because $k \geq 1$. Beyond the maximum of these cutoffs,

$$d_2(x^{(n)}, x) = \sum_{j=1}^{k} \left|x_j^{(n)} - x_j\right| < k\frac{\varepsilon}{k} = \varepsilon.$$

Thus a Cauchy sequence for either metric has a limit in that same metric and in the same space $\mathbb{R}^k$.

Conclusion. Both d_1 and d_2 are complete metrics on $\mathbb{R}^k$.

13.2 Exercise 13.2: Euclidean distance and convergence of coordinates

What we must show. Prove both inequalities in Lemma 13.3(1), then prove that a sequence in $\mathbb{R}^k$ converges if and only if every coordinate sequence converges in $\mathbb{R}$.

(a) Compare one summand, the sum, and its largest summand.

For $x, y \in \mathbb{R}^k$, let $M = \max_{1 \leq i \leq k} |x_i - y_i|$. Every square is nonnegative, so for each j,

$$|x_j - y_j|^2 \leq \sum_{i=1}^{k} |x_i - y_i|^2 \leq kM^2.$$

The first inequality retains one summand and discards only nonnegative summands. The second replaces each of the k summands by the upper bound M^2. Taking nonnegative square roots, which preserves order, gives

$$\boxed{|x_j - y_j| \leq d(x, y) \leq \sqrt{k} \max_{1 \leq i \leq k} |x_i - y_i|\,.}$$

(b), forward direction: Euclidean convergence gives coordinate convergence.

Suppose $x^{(n)} \to x$ in Euclidean distance. Given $\varepsilon > 0$, there is N such that $d(x^{(n)}, x) < \varepsilon$ for $n > N$. The first inequality in (a) then gives $\left|x_j^{(n)} - x_j\right| < \varepsilon$ for every j and $n > N$. Thus each $x_j^{(n)} \to x_j$ in $\mathbb{R}$.

(b), reverse direction: select one cutoff for all the coordinates.

Suppose $x_j^{(n)} \to x_j \in \mathbb{R}$ for $j = 1, \ldots, k$, and define $x = (x_1, \ldots, x_k)$. Given $\varepsilon > 0$, choose N_j such that

$$n > N_j \quad \Longrightarrow \quad \left|x_j^{(n)} - x_j\right| < \frac{\varepsilon}{\sqrt{k}}.$$

Set $N = \max_j N_j$. For $n > N$, the maximum of all coordinate errors is smaller than $\varepsilon/\sqrt{k}$. The second inequality in (a) gives $d(x^{(n)}, x) < \varepsilon$.

Conclusion. The distance bounds are proved, and coordinate convergence is equivalent to Euclidean convergence.

13.3 Exercise 13.3: The supremum metric on bounded sequences

What we must show. Let B be the set of all bounded real sequences. Prove that

$$d(x, y) = \sup_{j \in \mathbb{N}} |x_j - y_j|$$

is a metric on B, and decide whether the analogous infinite sum defines a metric on all of B.

(a) First verify that the supremum is a finite real number.

For $x, y \in B$, choose finite nonnegative bounds C_x, C_y with $|x_j| \le C_x$ and $|y_j| \le C_y$ for all j. Then

$$0 \le |x_j - y_j| \le |x_j| + |y_j| \le C_x + C_y.$$

The set of coordinate differences is nonempty and bounded above, so its supremum exists and belongs to $[0, C_x + C_y]$.

(a) Check the zeros, symmetry, and triangle inequality.

If $x = y$, every difference is zero and so is the supremum. If $d(x, y) = 0$, then each $0 \le |x_j - y_j| \le 0$, implying $x_j = y_j$ for all j, hence $x = y$. Symmetry follows coordinate by coordinate from symmetry of absolute value. Finally, for every j,

$$|x_j - z_j| \le |x_j - y_j| + |y_j - z_j| \le d(x, y) + d(y, z).$$

The final expression is an upper bound for all coordinate differences between x and z. Their least upper bound is thus at most that expression. This is the triangle inequality for d.

(b) A sum can be infinite even when both sequences are bounded.

Take $x = (0, 0, 0, \ldots)$ and $y = (1, 1, 1, \ldots)$. Both belong to B, but

$$\sum_{j=1}^{\infty} |x_j - y_j| = \sum_{j=1}^{\infty} 1 = +\infty,$$

since its nth partial sum is n. A metric in Definition 13.1 must give a finite real distance between each pair of points.

Conclusion. The supremum formula is a metric on B. The sum formula is not a metric on all of B, because it need not be finite.

13.4 Exercise 13.4: Unions and finite intersections of open sets

What we must show. Prove that arbitrary unions and finite intersections of open subsets of a metric space (S, d) are open.

(iii) For a point of a union, use one set that contains it.

Let $U = \bigcup_{\lambda\in\Lambda} U_\lambda$, where each U_λ is open. If $x \in U$, there is a λ_0 with $x \in U_{\lambda_0}$. Since that set is open, some $r > 0$ satisfies $B(x, r) \subseteq U_{\lambda_0} \subseteq U$. Thus every point of U is interior, which says U is open. If the collection is empty, its union is the empty set. That set is open because there is no point for which the interior condition could fail.

(iv) For a point of a finite intersection, take the smallest radius.

Let $V = U_1 \cap \cdots \cap U_m$, where $m \geq 1$ and all the U_i are open. For $x \in V$, choose $r_i > 0$ with $B(x, r_i) \subseteq U_i$. The number $r = \min\{r_1, \ldots, r_m\}$ is positive because this is a finite nonempty set of positive numbers. If $d(y, x) < r$, then $d(y, x) < r_i$ for every i, so $y \in U_i$ for every i. Hence $B(x, r) \subseteq V$, proving openness. The intersection of no subsets of S is S, which is open: every ball with center in S is a subset of S.

An infinite intersection need not be open: in $\mathbb{R}$, $\bigcap_{n\geq 1}(-1/n, 1/n) = \{0\}$, and no positive-radius interval about 0 fits in $\{0\}$. This explains why the intersection statement requires finiteness.

Conclusion. Arbitrary unions and finite intersections preserve openness.

13.5 Exercise 13.5: De Morgan's law and intersections of closed sets

What we must show. Verify the stated complement identity and deduce that an arbitrary intersection of closed subsets of S is closed.

(a) Prove equality by testing membership of a point.

For any $x \in S$, the following statements are equivalent:

$$\begin{aligned} x \in \bigcap_{U\in\mathcal{U}} (S \setminus U) \quad &\iff \quad x \notin U \text{ for every } U \in \mathcal{U} \\ &\iff \quad x \notin \bigcup_{U\in\mathcal{U}} U \\ &\iff \quad x \in S \setminus \bigcup_{U\in\mathcal{U}} U. \end{aligned}$$

Since every point has the same membership on the two sides, the sets are equal. For the empty collection, the left side is S and the right side is $S \setminus \varnothing = S$, so the convention is consistent.

(b) Take complements, then apply the open-union result.

Let F_λ be closed subsets of S, and put $F = \bigcap_{\lambda\in\Lambda} F_\lambda$. A point fails to belong to this intersection exactly when it fails to belong to at least one F_λ. Thus

$$S \setminus F = \bigcup_{\lambda\in\Lambda} (S \setminus F_\lambda).$$

Each complement on the right is open by the definition of a closed set. Their union is open by Exercise 13.4, so F is closed by that same definition. If Λ is empty, $F = S$ and its complement is empty and open.

Conclusion. The complement identity holds, and every intersection of closed sets is closed.

13.6 Exercise 13.6: Four equivalent descriptions involving closure

What we must show. Prove all four assertions of Proposition 13.9 for a subset E of an arbitrary metric space (S, d).

Plan. First derive a ball characterization directly from the definition of closure as an intersection of closed supersets. Then obtain the sequence characterizations without assuming them.

Preparation: closure is a closed superset, and a ball detects its points.

The family of closed sets containing E is nonempty, because S belongs to it. Every set in that family contains E; therefore their intersection $\overline{E}$ contains E. Exercise 13.5 shows that the intersection is closed.

For any $x \in S$, we claim

$$x \in \overline{E} \quad \Longleftrightarrow \quad B(x, r) \cap E \neq \varnothing \text{ for every } r > 0. \tag{*}$$

If some ball $B(x, r)$ misses E, then its complement is a closed set containing E, since balls are open. This closed superset excludes x, so $x \notin \overline{E}$. Conversely, if $x \notin \overline{E}$, then at least one closed set C containing E excludes x; otherwise x would belong to their intersection. The open set $S \setminus C$ contains x, so it contains some $B(x, r)$, and that ball misses E. These two implications prove (*).

(a) A set is closed exactly when it equals its closure.

If E is closed, it is one of the closed supersets in the defining intersection, so $\overline{E} \subseteq E$. The reverse inclusion was already proved, giving equality. If $E = \overline{E}$, then E is closed because $\overline{E}$ was proved closed. This proves both directions.

(c) Closure points are exactly limits of sequences of points of E.

We establish (c) before (b) so that there is no circular argument. If $x \in \overline{E}$, apply (*) with radius $1/n$ to choose $x_n \in E$ with $d(x_n, x) < 1/n$. For any $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. If $n > N$, then $d(x_n, x) < 1/n < \varepsilon$. Hence $x_n \to x$.

Conversely, if $x_n \in E$ and $x_n \to x \in S$, fix $r > 0$. Convergence gives some index n with $d(x_n, x) < r$, so $x_n \in B(x, r) \cap E$. Every ball about x thus meets E, and (*) gives $x \in \overline{E}$. For $E = \varnothing$, no such sequence exists and its closure is empty, so the equivalence still holds.

(b) Closed sets are exactly those containing limits of their convergent sequences.

If E is closed and $x_n \in E$ tends to a point $x \in S$, part (c) gives $x \in \overline{E}$, and part (a) gives $\overline{E} = E$, so $x \in E$. Conversely, suppose E contains every such limit. For every $x \in \overline{E}$, part (c) supplies a sequence in E converging to x, so the assumption gives $x \in E$. Thus $\overline{E} \subseteq E$. Combined with the always-valid inclusion $E \subseteq \overline{E}$, this gives equality, and part (a) makes E closed.

(d) Boundary points see both the set and its complement.

By definition, the boundary is $\overline{E} \setminus E^{\circ}$. A point x lies outside E° exactly when no ball about x is contained in E. Indeed, the existence of such a ball would also put x in E, since its center belongs to the ball, and would make x interior. Saying no ball is contained in E is equivalent to saying every

ball meets $S \setminus E$. By (*) applied to this complement,

$$x \notin E^\circ \quad \Longleftrightarrow \quad x \in \overline{S \setminus E}.$$

Consequently

$$\overline{E} \setminus E^\circ = \overline{E} \cap \overline{S \setminus E},$$

which is precisely the assertion in (d).

Conclusion. All four claims follow from the closure definition and the ball criterion (*), with no sequence characterization assumed in advance.

13.7 Exercise 13.7: Decomposing an open real set into disjoint intervals

What we must show. Express every open $U \subseteq \mathbb{R}$ as a finite or countably infinite disjoint union of open intervals. Unbounded intervals are allowed; the empty set is the union of an empty family.

1. Group points that have no gap of U between them.

For $x, y \in U$, write $x \sim y$ when the whole closed segment with endpoints x, y lies in U. This relation is reflexive, because $[x, x] = \{x\} \subseteq U$, and symmetric because the segment is unchanged when its endpoints are interchanged. For transitivity, if $x \sim y$ and $y \sim z$, every point between x and z lies in the union of the segments joining x to y and y to z. That union is contained in U. Thus $x \sim z$, so $\sim$ is an equivalence relation. Its classes partition U into nonempty pairwise disjoint sets.

2. Show that each class is an interval and is open.

Let I be a class, and let $p < q$ belong to it. The equivalence relation gives $[p, q] \subseteq U$. For $p < t < q$, the segment $[p, t]$ is contained in $[p, q]$, so $p \sim t$ and $t \in I$. Thus every point between two points of I belongs to I.

For $z \in I$, openness of U gives $r > 0$ with $(z - r, z + r) \subseteq U$. Every t in that interval has its segment to z inside the interval, hence inside U. Therefore $t \sim z$ and $t \in I$. We have proved $(z - r, z + r) \subseteq I$, so I is itself open.

3. Identify the endpoints, including infinite endpoints.

Let $a = \inf I$ and $b = \sup I$ in the extended real sense; use $a = -\infty$ or $b = +\infty$ if the corresponding bound does not exist in $\mathbb{R}$. Since I is nonempty and open, it contains an interval around one of its points, and $a < b$. For $a < t < b$, the definitions of infimum and supremum give $p, q \in I$ with $p < t < q$; otherwise t would be a lower or upper bound contradicting $a < t < b$. The interval property puts $t \in I$. Conversely, a finite endpoint cannot lie in I: openness there would supply a point of I still smaller than a or still larger than b. Hence $I = (a, b)$, with the usual interpretations $(a, +\infty)$, $(-\infty, b)$, or $\mathbb{R}$.

4. Prove that there are at most countably many classes.

Fix an enumeration $q_1, q_2, \ldots$ of $\mathbb{Q}$. Each nonempty open interval I contains a rational by density. Assign to I the least index $m(I)$ for which $q_{m(I)} \in I$. This least index exists by the least-element property of $\mathbb{N}$. Distinct classes have distinct assigned indices, because they are disjoint and cannot contain the same rational. Thus the classes inject into $\mathbb{N}$. If there are infinitely many, list their assigned indices in increasing order and use that order to list the classes; every assigned index appears, since only finitely many positive integers precede it.

Conclusion. The equivalence classes give the required disjoint interval decomposition. The family may be empty, finite, or countably infinite, and intervals may be unbounded.

13.8 Exercise 13.8: Intervals, balls, half-spaces, and examples that are neither

What we must show. Verify the assertions in the first paragraph of Example 3 and all the assertions in Example 4. In the interval assertions assume $a < b$; in the ball assertions assume $r > 0$ and $k \geq 1$. We address the degenerate cases separately below.

(a) Open and closed intervals in the real line.

For $x \in (a, b)$, let $\rho = \min\{x - a, b - x\} > 0$. If $|y - x| < \rho$, then $a < y < b$. Thus (a, b) is open. The complement of $[a, b]$ is $(-\infty, a) \cup (b, +\infty)$. Each ray is open: for a point strictly on its allowed side, half its distance to the endpoint provides a ball staying on that side. Their union is open, so $[a, b]$ is closed.

Every $x \in (a, b)$ is interior to $[a, b]$ by the first estimate. The endpoint a is not interior: every ball about it contains $a - \rho/2 < a$ for its radius $\rho > 0$. Similarly b is not interior because $b + \rho/2$ is outside $[a, b]$. Hence $[a, b]^\circ = (a, b)$.

The closed set $[a, b]$ contains (a, b), so $\overline{(a, b)} \subseteq [a, b]$. Every ball about a meets (a, b): choose a displacement $0 < t < \min\{\rho, b - a\}$ and use $a + t$. The point $b - t$ does the same for b. The ball criterion of Exercise 13.6 therefore puts both endpoints in the closure. Interior points already belong to the set. Thus $\overline{(a, b)} = [a, b] = \overline{[a, b]}$. Subtracting the interior gives boundary $\{a, b\}$ for both intervals.

If $a = b$, the open interval is empty and has empty closure and boundary. The closed interval is the singleton $\{a\}$, which is closed, has empty interior, and has boundary $\{a\}$. This is why the boundary assertion for both intervals presumes $a < b$.

(b), balls: openness, closedness, and the possible boundary points.

For a center $c \in \mathbb{R}^k$, put

$$D = \{x : d(x, c) < r\}, \qquad K = \{x : d(x, c) \leq r\}.$$

The opening ball argument of this section shows D is open. If $x \notin K$, let $\rho = d(x, c) - r > 0$. For $d(y, x) < \rho$, the triangle inequality rearranges to

$$d(y, c) \geq d(x, c) - d(x, y) > d(x, c) - \rho = r.$$

Thus a ball about x stays outside K, so the complement of K is open and K is closed.

If $d(x, c) < r$, then x is interior to both sets, using the radius $r - d(x, c)$. If $d(x, c) > r$, the preceding complement ball shows x lies outside the closure of either set. Only points with $d(x, c) = r$ can therefore be boundary points.

(b), spheres: every point on the sphere really is a boundary point.

Let $d(x, c) = r > 0$, and let $\varepsilon > 0$. Choose $0 < t < \min\{1, \varepsilon/r\}$. Define

$$x_- = c + (1 - t)(x - c), \qquad x_+ = c + (1 + t)(x - c).$$

The Euclidean distance formula gives

$$d(x_-, c) = (1-t)r < r, \quad d(x_+, c) = (1+t)r > r, \quad d(x_-, x) = d(x_+, x) = tr < \varepsilon.$$

Thus every ball about x meets D (at x_-) and the complement of K (at x_+). It meets both either set and its complement, so Exercise 13.6(d) puts x on both boundaries. Consequently both boundaries are exactly $\{x : d(x, c) = r\}$. This radial argument uses Euclidean space; the same sphere assertion is not valid for all metrics. If $r = 0$, the open ball is empty with empty boundary, while the closed ball is the singleton $\{c\}$ with boundary $\{c\}$.

(b), strict coordinate inequalities give open half-spaces and quadrants.

In $\mathbb{R}^2$, consider $H = \{x : x_1 > 0\}$ and $Q = \{x : x_1 > 0,\ x_2 > 0\}$. For $x \in H$, choose $\rho = x_1/2 > 0$. Since $|y_1 - x_1| \le d(y, x)$, every y in $B(x, \rho)$ satisfies $y_1 > x_1 - \rho = x_1/2 > 0$. Thus H is open. For $x \in Q$, choose $\rho = \min\{x_1, x_2\}/2$. The same coordinate estimate, separately for $j = 1, 2$, gives $y_j > x_j - \rho \ge x_j/2 > 0$. Hence Q is open.

For $H_0 = \{x : x_1 \ge 0\}$, its complement is $\{x : x_1 < 0\}$, an open half-space by the same radius argument with $-x_1$ in place of x_1. Thus H_0 is closed. For $Q_0 = \{x : x_1 \ge 0,\ x_2 \ge 0\}$, the complement is $\{x : x_1 < 0\} \cup \{x : x_2 < 0\}$, a union of open half-spaces. Thus Q_0 is closed as asserted.

(b), the half-open interval is neither open nor closed.

The point $0 \in [0, 1)$ is not interior, since every ball about 0 contains negative points. Hence $[0, 1)$ is not open. The sequence $1 - 1/n$ belongs to $[0, 1)$ and tends to 1, which is outside the set. The closed-set sequence criterion of Exercise 13.6(b) shows it is not closed.

(b), the mixed quadrant is neither open nor closed.

Let $M = \{x : x_1 > 0,\ x_2 \ge 0\}$. The point $(1, 0)$ belongs to M, but every ball of radius $\rho > 0$ about it contains $(1, -\rho/2)$, outside M. Thus M is not open. The sequence $(1/n, 0)$ lies in M and tends in Euclidean distance to $(0, 0)$, outside M. It follows that M is not closed either.

Conclusion. All the interval, Euclidean ball, half-space, quadrant, and mixed examples have the stated properties, with the positive radius and nondegenerate interval conditions made explicit.

13.9 Exercise 13.9: Three closures in the real line

What we must show. Find the closures of $\{1/n : n \in \mathbb{N}\}$, $\mathbb{Q}$, and $\{q \in \mathbb{Q} : q^2 < 2\}$ in $\mathbb{R}$.

(a) Include the reciprocal terms and their one additional limit.

Let $A = \{1/n : n \in \mathbb{N}\}$. Every point of A belongs to its closure, and $1/n \to 0$ gives $0 \in \overline{A}$ by Exercise 13.6(c). We show no other point belongs. If $x < 0$, the ball of radius $-x/2$ about x is negative and misses A. If $x > 1$, the radius $(x-1)/2$ gives a ball wholly above 1, which also misses A.

It remains to exclude $0 < x \le 1$ with $x \notin A$. Choose N so large that $1/N < x/2$. All terms $1/n$ for $n \ge N$ are then below $x/2$. The finite set $\{1, 1/2, \ldots, 1/(N-1)\}$ has no member equal to x. Its finitely many positive distances from x have a positive minimum η. Here $N \ge 2$ because $1/N < x/2 \le 1/2$. A ball about x of radius $\min\{x/2, \eta\}/2$ misses the finite set and all the remaining reciprocals. The ball criterion excludes x. Therefore

$$\boxed{\overline{A} = A \cup \{0\}.}$$

(b) Density fills the whole real line.

For any $x \in \mathbb{R}$ and $r > 0$, the interval $(x - r, x + r)$ contains a rational by density of $\mathbb{Q}$. Hence every real x satisfies the closure ball criterion, giving $\boxed{\overline{\mathbb{Q}} = \mathbb{R}}$.

(c) The rational points between the two square-root endpoints.

For real q, the inequality $q^2 < 2$ is equivalent to $-\sqrt{2} < q < \sqrt{2}$. Thus the given set is $C = \mathbb{Q} \cap (-\sqrt{2}, \sqrt{2})$. If $-\sqrt{2} < x < \sqrt{2}$, then every neighborhood of x contains a smaller nonempty interval lying inside $(-\sqrt{2}, \sqrt{2})$; density supplies a rational in that interval. For $x = \sqrt{2}$, an interval $(\sqrt{2} - t, \sqrt{2})$ with $0 < t < \min\{r, 2\sqrt{2}\}$ lies inside the radius-r neighborhood and inside the allowed range, and contains a rational. The interval $(-\sqrt{2}, -\sqrt{2} + t)$ proves the same statement for $x = -\sqrt{2}$. Therefore the whole closed interval belongs to $\overline{C}$.

If $x > \sqrt{2}$, take radius $(x - \sqrt{2})/2$; if $x < -\sqrt{2}$, take radius $(-\sqrt{2} - x)/2$. The resulting ball misses the closed interval and hence misses C. No outside point belongs to the closure, so

$$\boxed{\overline{C} = [-\sqrt{2}, \sqrt{2}].}$$

Conclusion. The closures are respectively $\{0\} \cup \{1/n : n \in \mathbb{N}\}$, $\mathbb{R}$, and $[-\sqrt{2}, \sqrt{2}]$.

13.10 Exercise 13.10: Three sets with no interior points

What we must show. Prove that the reciprocal set, the rationals, and the Cantor set all have empty interior in $\mathbb{R}$.

(a) A rational-only set cannot contain a real interval.

Every reciprocal $1/n$ is rational. Every open interval contains an irrational number. For completeness, to find one in (a, b), choose a rational q in $(a - \sqrt{2}, b - \sqrt{2})$ by rational density; then $q + \sqrt{2} \in (a, b)$ is irrational, since its being rational would make $\sqrt{2} = (q + \sqrt{2}) - q$ rational. Thus every ball about any point of the reciprocal set contains a point outside that set. No point of the set is interior.

(b) Apply the same irrational-density argument to $\mathbb{Q}$.

For any $q \in \mathbb{Q}$ and $r > 0$, the preceding argument gives an irrational in $(q - r, q + r)$. That ball is not a subset of $\mathbb{Q}$. Thus $\mathbb{Q}$ also has no interior points. Notice that dense closure does not imply nonempty interior: Exercise 13.9 gave $\overline{\mathbb{Q}} = \mathbb{R}$.

(c) The intervals remaining at late Cantor stages are too short.

Use the numbering in Example 5: $F_1 = [0, 1]$, and F_n is the union of 2^{n-1} disjoint closed intervals, each of length $3^{-(n-1)}$. The Cantor set is $F = \bigcap_{n \geq 1} F_n$. Suppose, for contradiction, that $x \in F^\circ$. Then some real interval $(x - r, x + r)$, with $r > 0$, lies in F. The closed interval $J = [x - r/2, x + r/2]$ has length r and also lies in F, hence in every F_n.

Choose n so large that $3^{-(n-1)} < r$. The interval J would have to lie in one of the component intervals of F_n. Indeed, if it met two distinct components, it would also contain the points of the nonempty removed gap between those components, because it contains every point between any two of its points. That would contradict $J \subseteq F_n$. But an interval of length r cannot be contained in an interval of smaller length $3^{-(n-1)}$. This is a contradiction.

Conclusion. All three interiors are empty. The Cantor proof uses the explicit shrinking intervals of its construction.

13.11 Exercise 13.11: Compactness and subsequences in Euclidean space

What we must show. Show that $E \subseteq \mathbb{R}^k$ is compact if and only if every sequence of points of E has a subsequence converging to a point of E.

Plan. Use Heine–Borel and Bolzano–Weierstrass. For the converse, show that the stated subsequence property forces both boundedness and closedness.

Preparation: uniqueness and inheritance of metric limits.

If $x_n \to x$ and $x_n \to y$, then $d(x,y) \le d(x,x_n) + d(x_n,y) \to 0$, so $d(x,y) = 0$ and $x = y$. Every subsequence of a convergent sequence has the same limit: strictly increasing selection indices n_j satisfy $n_j \ge j$, so every eventual distance estimate for the original sequence also holds for all sufficiently late selected terms. Also, a convergent sequence in $\mathbb{R}^k$ is bounded: its tail lies within distance 1 of its limit x, and for each coordinate the triangle inequality gives $\left|x_j^{(n)}\right| \le |x_j| + 1$ on that tail. A finite maximum accommodates the omitted initial coordinates.

Forward direction: combine boundedness, subsequences, and closedness.

If E is compact, Heine–Borel (Theorem 13.12) says it is closed and bounded. Any sequence in E is then bounded. Bolzano–Weierstrass (Theorem 13.5) gives a subsequence tending to some point of $\mathbb{R}^k$. Since all its terms lie in the closed set E, Exercise 13.6(b) puts that limit in E.

Reverse direction, boundedness: unbounded points would escape every subsequence.

Assume the stated subsequence property. If E were unbounded, then for each $n \in \mathbb{N}$ we could choose $x^{(n)} \in E$ with

$$\max_{1 \le j \le k} \left|x_j^{(n)}\right| > n.$$

For any subsequence $x^{(n_m)}$, the selected indices satisfy $n_m \ge m$, so its coordinate maximum exceeds m. It is therefore unbounded: for any proposed finite bound C, take $m > C$. It cannot converge, since a convergent sequence was just shown to be bounded. This contradicts the assumed property. Thus E is bounded.

Reverse direction, closedness: recover every closure point.

Let $x \in \overline{E}$. By Exercise 13.6(c), choose a sequence $x_n \in E$ converging to x. The assumed property supplies a subsequence $x_{n_m} \to y$ for some $y \in E$. Inheritance gives $x_{n_m} \to x$ as well, and uniqueness of the limit gives $x = y \in E$. Thus $\overline{E} \subseteq E$. The reverse inclusion always holds, so Exercise 13.6(a) makes E closed. Having proved both closedness and boundedness, Heine–Borel gives compactness.

If $E = \varnothing$, it is compact because the empty subfamily is a finite subcover of every cover of it. There is no sequence of points of E, so the sequential condition is vacuously true.

Conclusion. The two conditions are equivalent for all subsets of $\mathbb{R}^k$, including the empty set.

13.12 Exercise 13.12: Closed subsets and finite unions of compact sets

What we must show. In any metric space (S, d), prove that a closed subset of a compact set is compact, and that a finite union of compact sets is compact.

Plan. Work directly with open covers, so neither result depends on the finite-dimensional Heine–Borel theorem.

(a) Add one open set to cover the rest of the larger compact set.

Suppose $E \subseteq F$, E is closed in S, and F is compact. Let $\mathcal{U}$ be any family of open subsets of S covering E. The set $V = S \setminus E$ is open, and $\mathcal{U} \cup \{V\}$ covers F: a point of F either belongs to E and is covered by $\mathcal{U}$, or lies in V. Compactness of F gives a finite subfamily covering F. Remove V if it was selected. The remaining sets still cover E, since no point of E belongs to V. They are a finite subfamily of $\mathcal{U}$, so E is compact.

If "closed subset of F" is interpreted as closed relative to F, the same conclusion holds: relative closedness means $F \setminus E = F \cap V$ for some open $V \subseteq S$. Use that V in the preceding cover argument; it still misses E and covers every point of $F \setminus E$.

(b) Take a finite subcover for each compact piece.

Let $K = K_1 \cup \cdots \cup K_m$ with each K_i compact, and let $\mathcal{U}$ be an open cover of K. It also covers each K_i. For each i, select a finite subfamily $\mathcal{U}_i \subseteq \mathcal{U}$ covering K_i. The family $\mathcal{U}_1 \cup \cdots \cup \mathcal{U}_m$ is finite, since a finite union of finite sets is finite, and it covers every piece, hence their union K. This proves compactness. When $m = 0$, the union is empty and is compact by the empty-subcover argument.

Conclusion. Both permanence properties hold in every metric space.

13.13 Exercise 13.13: A compact real set contains its extreme bounds

What we must show. For a nonempty compact $E \subseteq \mathbb{R}$, show $\sup E \in E$ and $\inf E \in E$.

1. Ensure the bounds are finite and use the approximation property.

Heine–Borel makes E closed and bounded. Since it is also nonempty, completeness of $\mathbb{R}$ supplies finite real numbers $M = \sup E$ and $m = \inf E$. For every $n \in \mathbb{N}$, $M - 1/n$ is strictly below the least upper bound, so it cannot be an upper bound. Choose $u_n \in E$ with

$$M - \frac{1}{n} < u_n \leq M.$$

Similarly, $m + 1/n$ is strictly above the greatest lower bound, so it cannot be a lower bound. Choose $v_n \in E$ with

$$m \leq v_n < m + \frac{1}{n}.$$

2. Check convergence, then use closedness.

The first inequalities give $|u_n - M| < 1/n$, so $u_n \to M$. The second give $|v_n - m| < 1/n$, so $v_n \to m$. Both sequences lie in E. Closedness and Exercise 13.6(b) therefore put both limits in E.

Conclusion. The supremum and infimum are attained: E has a maximum and a minimum, namely M and m.

13.14 Exercise 13.14: The diameter of a compact set is attained

What we must show. For nonempty compact $E \subseteq \mathbb{R}^k$, put

$$\delta = \sup\{d(x, y) : x, y \in E\}.$$

Find $x^0, y^0 \in E$ such that $d(x^0, y^0) = \delta$.

Plan. Approximate the supremum by pairs, select subsequences for both entries of the pairs, and justify passage to the limit using the triangle inequality.

1. Verify that the distance supremum is a finite real number.

Heine–Borel makes E bounded. Choose $C \geq 0$ so that every coordinate of every $x \in E$ has absolute value at most C. For $x, y \in E$, each $|x_j - y_j| \leq 2C$, so $d(x, y) \leq 2C\sqrt{k}$ by Exercise 13.2(a). The set of distances is nonempty because E is nonempty, and contains $d(x, x) = 0$. Its supremum therefore exists with $0 \leq \delta \leq 2C\sqrt{k}$.

2. Choose pairs whose distances approach the supremum.

For each $n \in \mathbb{N}$, the real number $\delta - 1/n$ is not an upper bound of the distance set. Hence choose $x_n, y_n \in E$ such that

$$\delta - \frac{1}{n} < d(x_n, y_n) \leq \delta.$$

This works even when $\delta = 0$; any pair then has distance $0 > \delta - 1/n$. The displayed error bound gives $d(x_n, y_n) \to \delta$.

3. Pass to a common subsequence for the two endpoints.

By Exercise 13.11, some subsequence x_{n_j} tends to a point $x^0 \in E$. Apply the same result to the sequence y_{n_j} in E to find further indices j_ℓ with $y_{n_{j_\ell}} \to y^0 \in E$. The matching $x_{n_{j_\ell}}$ still tends to x^0, since a subsequence inherits a limit. This further selection is also a subsequence of the original sequence of pairs because the composition of two increasing index selections is increasing.

4. Prove that distances pass to the limit.

Two uses of the triangle inequality give

$$d(x, y) \leq d(x, x^0) + d(x^0, y^0) + d(y^0, y),$$
$$d(x^0, y^0) \leq d(x^0, x) + d(x, y) + d(y, y^0).$$

Subtracting $d(x^0, y^0)$ in the first and $d(x, y)$ in the second combines into

$$\Big|d(x, y) - d(x^0, y^0)\Big| \leq d(x, x^0) + d(y, y^0).$$

At the selected pairs, the right side tends to zero. Thus $d(x_{n_{j_\ell}}, y_{n_{j_\ell}}) \to d(x^0, y^0)$. The same distances tend to δ, being a subsequence of the distance sequence in Step 2. Uniqueness of real limits gives $d(x^0, y^0) = \delta$.

Conclusion. Two points of E realize its diameter, including the zero-diameter case.

13.15 Exercise 13.15: A closed bounded set that is not compact

What we must show. In the bounded-sequence space (B, d) of Exercise 13.3, let

$$F = \left\{ x \in B : \sup_{j \in \mathbb{N}} |x_j| \leq 1 \right\}.$$

Show that F is closed and bounded, but not compact.

Plan. Verify closedness using an open complement. Then use the prescribed cover by open balls of radius 1, together with infinitely many points whose pairwise distance is 2.

(a), boundedness: use the zero sequence as the center.

The sequence $0 = (0, 0, \ldots)$ belongs to B. For $x \in F$,

$$d(x, 0) = \sup_j |x_j| \leq 1.$$

Thus F lies in the closed metric ball with center 0 and radius 1, which is exactly the boundedness condition stated in the exercise.

(a), closedness: exhibit a ball about each point outside F.

If $x \in B \setminus F$, then $\sup_j |x_j| > 1$. At least one coordinate, say j_0, satisfies $|x_{j_0}| > 1$; otherwise 1 would be an upper bound for all these absolute values and hence for their supremum. Set $\rho = (|x_{j_0}| - 1)/2 > 0$. If $d(x, y) < \rho$, the real triangle inequality gives

$$|y_{j_0}| \geq |x_{j_0}| - |x_{j_0} - y_{j_0}| > |x_{j_0}| - \rho = \frac{|x_{j_0}| + 1}{2} > 1.$$

Such a y lies outside F. Hence a ball about each point of the complement stays in the complement, proving it open and F closed in B.

(b), the open cover: give every point its radius-one ball.

For $x \in F$, define $U(x) = \{y \in B : d(y, x) < 1\}$. These sets are open in B by the general ball fact at the start of this section. They cover F, since $d(x, x) = 0 < 1$ puts each $x \in F$ in its own $U(x)$.

(b), separated points: compute the distance exactly.

For each $n \in \mathbb{N}$, let $x^{(n)}$ have coordinate -1 at index n and coordinate 1 at every other index. Every one of its coordinates has absolute value 1, so $x^{(n)} \in F$. If $m \neq n$, the two sequences differ by 2 in absolute value at coordinates m and n, and agree at all other coordinates. Therefore

$$d(x^{(m)}, x^{(n)}) = 2.$$

No ball $U(x)$ can contain two distinct such points: if it contained both, then the metric triangle inequality would give

$$2 = d(x^{(m)}, x^{(n)}) \leq d(x^{(m)}, x) + d(x, x^{(n)}) < 1 + 1 = 2,$$

an impossibility. The strict inequalities in the open-ball definition are essential here.

(b), exclude every finite subcover.

A union of q members of this cover can contain at most q of the separated points, since each member contains at most one. But F contains the infinitely many distinct points $x^{(1)}, x^{(2)}, \ldots$. For example, among its first $q+1$ separated points at least one must be missed. Thus no finite subfamily covers F.

Conclusion. F is closed and bounded in the specified metric space, but this explicit open cover has no finite subcover, so F is not compact. Heine–Borel's closed-and-bounded criterion is specific to Euclidean space and does not apply to all metric spaces.